Exercise Sheet 3

Due: 10.11.26 9h00

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General Regulations.

- Submission is optional but recommended; tutors provide feedback. Exercises are not graded.
- Work in groups of two or three, preferably from the same tutorial group. Submit all solutions as a single combined PDF via the Übungsgruppenverwaltung, once per group, and include all group members' names.
- Solutions to tasks marked "solve yourself, without an AI agent" may be handwritten and scanned or typeset using LaTeX. Please ensure that scans are legible and not blurry.
- For computational tasks, include the relevant code, results and your interpretation in the PDF.
- Course material is available in the GitHub repository.

1 CT reconstruction

Solve parts (a), (b), (c) and (e) yourself, without an AI agent. Use an agent to write the code and implement your tests in part (d).

Computed tomography reconstructs an object's internal attenuation from X-rays passing through it at different angles. For a single ray, the Beer–Lambert law gives

$$I_1 = I_0 \exp\left(- \int_{\text{ray}} \mu(l)\, \mathrm{d}l \right),$$

where I_0 is the emitted and I_1 the detected intensity, and $\mu(l)$ is the attenuation per unit length. Approximate the object by p pixels, with constant attenuation w_k inside pixel k. For ray j, let X_{kj} be its path length through that pixel.

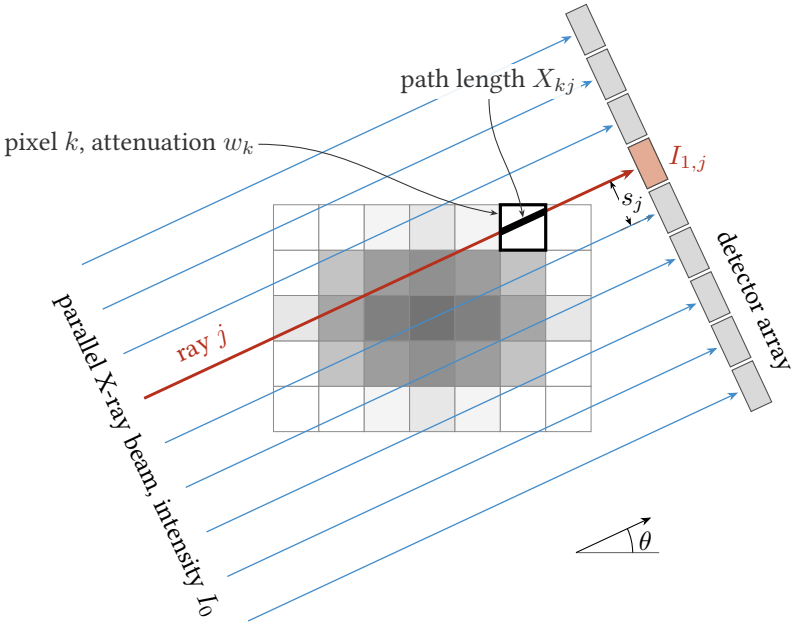


Figure 1: Schematic parallel-beam CT setup (fewer rays and pixels than in the dataset). At beam angle θ, parallel rays with intensity I_0 cross the pixelated object; ray j, at offset s_j, reaches its detector with intensity $I_{1,j}$. Inside pixel k it travels the path length X_{kj} (thick segment). Source and detector array rotate together to record all angles.

(a) Set up the inverse problem. Define $y_j = -\ln(I_{1,j}/I_0)$ and derive the noise-free relation between y_j, X_{kj} and w_k. Write your relation for n rays in matrix form, using $X \in \mathbb{R}^{p \times n}$, and state the

dimensions of $\mathbf{y}$ and $\mathbf{w}$. Which quantities are known and which are to be reconstructed? Why can the reconstruction be ambiguous when $p > n$?

(b) Interpret the design matrix. What does one column of X represent, and what does one row represent? Explain which can be displayed as an image of the object and which as an image with angle and detector position as axes.

The dataset contains a simulated scan of a 99×117 image, measured at 30 angles with 155 parallel rays per angle. Neighbouring rays are one pixel width apart. Arranged with one row per angle and one column per detector position, the measurements form a 30×155 image called the *sinogram*. The file `sheet03/data/sino.npy` contains the noisy measurements y_j; `sheet03/ct_data.py` provides the corresponding design matrix X of path lengths, in units of the pixel width. The data folder's `README.txt` describes the file contents and measurement geometry.

(c) Design tests before using an agent. When working with an AI agent, you remain responsible for checking its code. A plausible-looking reconstruction is not enough: useful tests need expectations justified independently of the implementation.

Before consulting an agent, propose at least one meaningful test for the reconstruction task in part (d). For each test, describe the input or setup, the expected result or property and why it should hold, and the kind of error it could reveal.

(d) Reconstruct the scan. Use an agent to visualize the sinogram and four selected columns and four selected rows of X. Reconstruct the image using ridge regression,

$$\min_{\mathbf{w}} \ \|\mathbf{y} - X^{\mathsf{T}}\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_2^2,$$

for regularization strengths on a logarithmic grid from $\lambda = 10^{-6}$ to $\lambda = 10^{4}$. Visualize the reconstructions and report their data residuals $n^{-1}\|\mathbf{y} - X^{\mathsf{T}}\widehat{\mathbf{w}}\|_2^2$.

Give the agent your test plan from part (c) and ask it to implement and run those tests on its code. Review whether the implemented tests check what you intended.

(e) Assess the reconstructions. Describe how the reconstructed image and its data residual change with λ. For which values of λ is the image dominated by amplified noise, and for which is it oversmoothed? Does a smaller residual establish that the image is closer to the unknown object?

2 Bayes decisions: signal or background?

Solve this exercise yourself, without an AI agent.

An imaging atmospheric Cherenkov telescope must decide which candidate events to retain for further analysis. Retaining background wastes limited storage and processing capacity; discarding a true gamma ray loses scientifically valuable data. For this simplified model, an event is either a gamma ray from the target (S, signal) or background (B), with probabilities $P(S) = 0.05$ and $P(B) = 0.95$. An analysis checks whether the reconstructed arrival direction is aligned with the target. Let D denote an aligned event, with

$$P(D \mid S) = 0.95, \qquad P(D \mid B) = 0.1.$$

(a) Posterior probability. Compute $P(S \mid D)$. Would you retain or discard the aligned event if both types of error have loss 1 and correct decisions have loss 0 (0–1 loss)?

(b) Changing the loss. Let $q = P(S \mid x)$ for an observed event x. Retaining background has loss $c_{\mathrm{FP}} > 0$; discarding signal has loss $c_{\mathrm{FN}} > 0$. The other two outcomes have zero loss. Write the conditional expected loss of each action and derive the optimal retain-or-discard rule in terms of q. Now assign loss 1 to wasted resources ($c_{\mathrm{FP}} = 1$) and loss 10 to lost signal ($c_{\mathrm{FN}} = 10$). Should you retain the aligned event from part (a)?

(c) Bayes risk. Let $\bar{D}$ denote a non-aligned event. Determine the optimal decision for non-aligned events under both loss functions. Compute the Bayes risk, i.e. the expected loss per event, of each optimal rule and compare it with the rules “always retain” and “always discard”. When does the alignment check help?